Assignment 4: AI-Based Fault Detection in Digital Circuits

1. GenAI Model Name:

ChatGPT (OpenAI)

2. Prompt:

Design a Full Adder circuit with a deliberately introduced fault. Use its truth table to identify the incorrect output, ask GenAI to diagnose the fault, and provide the corrected Boolean equations and gate-level circuit. Verify the corrected circuit using Logisim Evolution.

3. Response:

A Full Adder is a combinational circuit that adds three 1-bit inputs: A, B and Carry-in (Cin). It produces two outputs: Sum (S) and Carry-out (Cout).

Correct Boolean equations:

Sum = $A \oplus B \oplus Cin$

Cout = AB + ACin + BCin

Deliberately introduced fault:
The Carry-out circuit is incorrectly designed as:
Cout_faulty = AB + ACin

The term BCin is missing. Therefore, the faulty circuit may produce an incorrect carry output for some input combinations.

Truth table:

A	B	Cin	Correct Sum	Correct Cout	Faulty Cout
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	0	0
1	0	1	0	1	1
1	1	0	0	1	1
1	1	1	1	1	1

AI-based fault diagnosis:

The truth table shows a mismatch at input combination A = 0, B = 1 and Cin = 1. The correct Carry-out is 1, but the faulty circuit produces 0. The missing term is BCin.

Corrected equation:

Cout = AB + ACin + BCin

4. Verification by Logisim:

The faulty Full Adder circuit should first be built using XOR, AND and OR gates. Test all eight input combinations and compare the outputs with the correct truth table.

Then add the missing AND term BCin to the Carry-out logic and test all eight combinations again. The corrected circuit should produce the expected Sum and Carry-out values for every input combination.

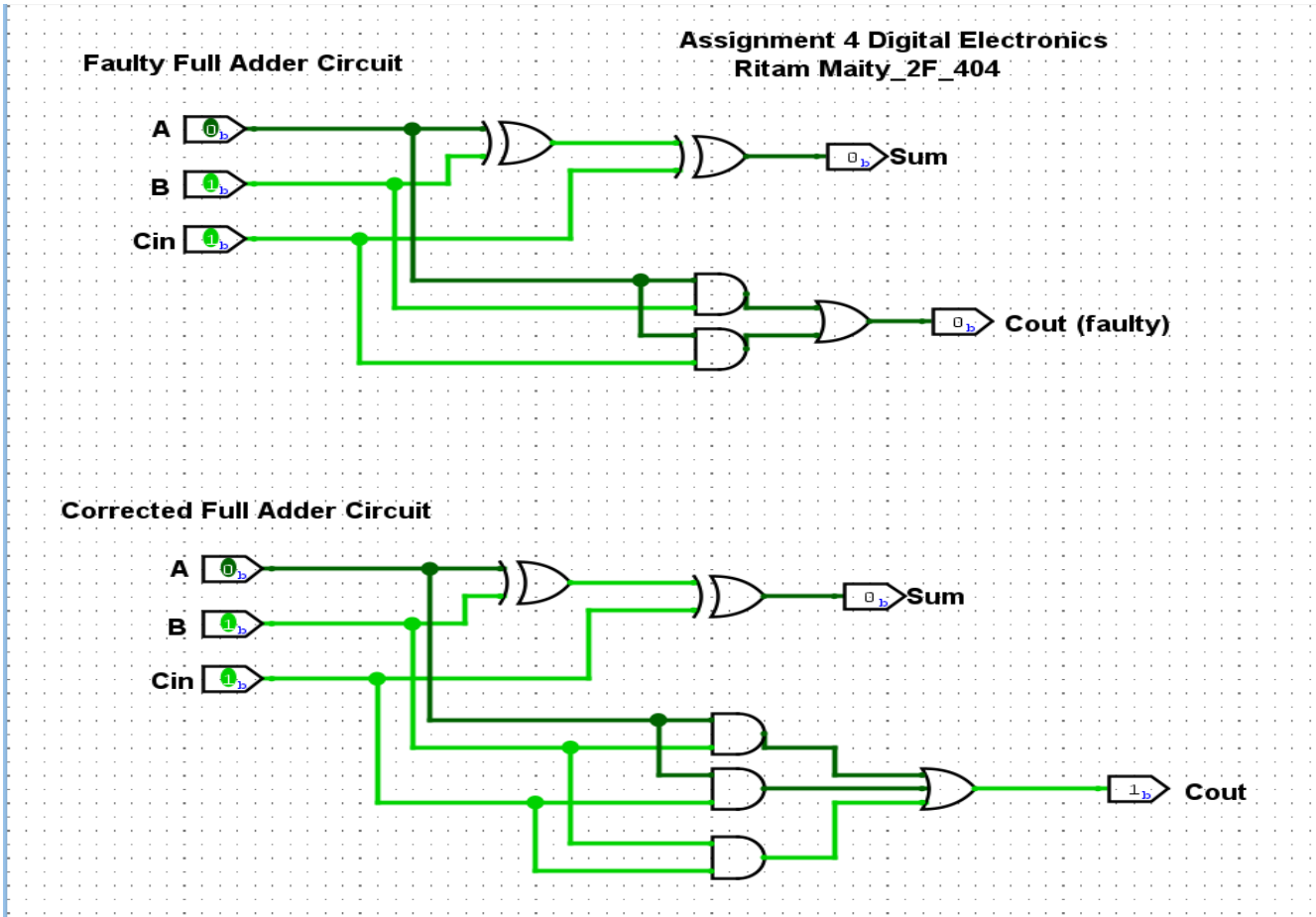
5. Circuit:

Faulty circuit:

- Sum = A XOR B XOR Cin.
- Carry-out = (A AND B) OR (A AND Cin).
- The BCin term is missing.

Corrected circuit:

- Connect A and B to an XOR gate.
- Connect the XOR output and Cin to another XOR gate to generate Sum.
- Use three AND gates to generate AB, ACin and BCin.
- Connect the three AND outputs to an OR gate to generate Carry-out.



7. Discussion

This assignment demonstrates how GenAI can help detect faults in digital circuits by comparing actual and expected truth tables. The missing BCin term caused the incorrect Carry-out. Adding this term corrected the circuit. Testing all input combinations helps confirm the correction.

8. GitHub Link

https://github.com/Ritam-Maity/Digital-Electronics-Lab